Measurement-shifting in the syntax:
A rational solution to an agreement-interpretation puzzle?

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1 Introduction

This work concerns FRACTION PARTITIVES (FPs), a nominal construction exemplified in (1). FPs which embed a plural DP are canonically understood to denote a subset of their complement whose cardinality is that of the complement multiplied by the fraction. Ergo, (1) could mean two out of six apples, three out of nine apples, etc. (see, e.g., Ionin et al. 2006; Wagiel 2021; Benbaji-Elhadad and Wehbe 2024).

(1) A third of the apples

For languages like American English, this understanding is fairly suitable. However, some languages exhibit an additional reading for FPs which has received very little attention in the literature. To illustrate, imagine a dingy studio apartment in Rome, Italy. Aerially, this apartment is an unequilateral hexagon, with two four-meter-wide walls and four one-meter-wide walls. Assuming the walls are of uniform height, the combined surface area of the apartment's walls is $12 \times h$ square meters (where h is the apartment's height). This is diagrammed in *fig.* 1.

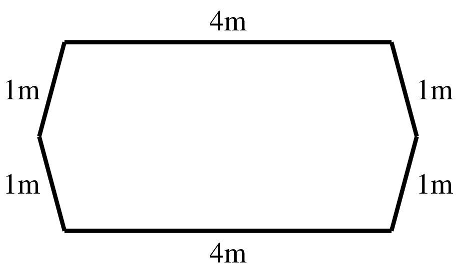


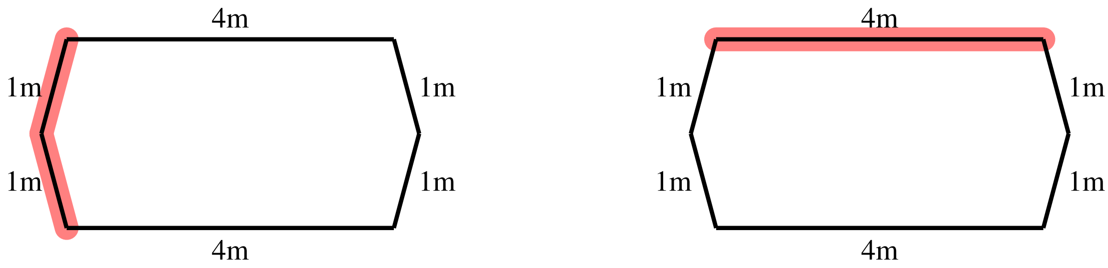
Figure 1: Floor plan of the studio apartment (not to scale)

Let's say this apartment has developed a mold problem—blight has completely covered one third of the walls. If the Italian-speaking tenant of this unit called their landlord and said (2), this would be compatible with two scenarios. First, it may be that two out of the six walls are completely covered in mold (*fig.* 2a); this interpretation, which I presented above, is what I call the COUNT reading. Alternatively, it could be that one of the large walls (whose surface area is a third of that of the combined walls) is completely covered in mold (*fig.* 2b); I dub this interpretation, which is unavailable in American English, the MEASURE reading.[1]

(2) la muffa ha coperto un terz-o del-le paret-i
DET.F.SG mold has.3SG covered.3SG one third-M.SG of-DET.F.PL wall-F.PL
'The mold has covered a third of the walls' [Italian]

What is even more interesting is that Italian grammar has an inherent way to circumvent this confusion. If a FP is in a position of agreement, agreeing elements can expone the features of either the fraction (in this case, *terzo* (masculine singular)) or the complement of the FP (*le pareti* (feminine plural)). If the denominator is agreed with, only the measure reading is available. If instead agreement is with the complement, only the count reading is available.

[1] My use of *unambgiuously* in *fig.* 2a, 2b should be clear; if the apartment layout was s.t. every member of the wall plurality was congruent (same size and shape), then there would be substantial overlap between the count and measure reading.



(a) Unambiguously COUNT context (b) Unambiguously MEASURE context

Figure 2: Two possible scenarios for mold coverage

(3) 'One third of the walls are covered by mold' [Italian]

a. (esatamente) un **terz-o** del-le paret-i è coperto di muffa
(exactly) one third-M.SG of-DET.F.PL wall-F.PL be.3SG covered.M.SG by mold
✓M, ×C

b. (esatamente) un terz-o del-**le** **paret-i** sono coperte di muffa
(exactly) one third-M.SG of-DET.F.PL wall-F.PL be.3PL covered.F.PL by mold
×M, ✓C

Of course, the measure interpretation is compatible with other scenarios as well; the only constraint is that the cumulative square meterage of the moldy region be $4h\ m^2$. It could be that the lower third of each wall is covered in mold, In an even more dire accomodation situation where there are only two walls, it could be that two thirds of one wall are covered in mold. I use the dichotomy in *fig.* 2a, 2b for expository purposes.

This general pattern extends to other kinds of count nouns besides just *walls*. (4) represents a case with an abstract noun like *film*.

(4) 'A third of the films were watched by John' [Italian]

a. **un terzo** de-i film è stato visto
one third of-M.PL film COP.3SG been.M.SG watched.M.SG
✓M, ×C

b. un terzo de-**i** **film** sono stati visti
one third of-M.PL film COP.3SG M.PL watched.M.PL
×M, ✓C

Furthermore, this is not a phenomenon unique to Italian; it can be observed to some extent in many Indo-European languages. French and Greek exhibit display exactly the Italian pattern.

(5) 'One third of the walls are covered by mold' [French]

a. **un tiers** des murs est couvert de moisissures
one third of.DET.M.PL wall.M.PL COP.3SG covered.M.SG by mold
✓M, ×C

b. un tiers **des** **murs** sont couverts de moisissures
one third of.DET.M.PL wall.M.PL COP.3PL covered.M.PL by mold
×M, ✓C

(6) 'One third of the pies were eaten' [Greek]

a. **ena trit-o** ton pit-on fagoθik-e
one third-N.SG DET.GEN.PL pie-GEN.PL eat.PASS.PRF.PST-3SG.PST
✓M, ×C

b. ena trit-o **ton** **pit-on** fagoθik-an
one third-N.SG DET.GEN.PL pie-GEN.PL eat.PASS.PRF.PST-3PL.PST
×M, ✓C

A similar pattern can be observed with the fraction partitives of Russian and German, albeit with the caveat that a count reading is still available when Agreement is with the denominator.

(7) 'One third of these[2] cakes were eaten' [Russian]

a. **(odna) tret'** étix pirog-ov byl-a s"eden-a
one third.F.SG this.GEN.PL cake-GEN.PL be-F.SG eaten-F.SG
✓M, ✓C

b. (odna) tret' **étix** **pirog-ov** byl-i s"eden-i
one third.F.SG this.GEN.PL cake-GEN.PL be-PL eaten-PL
×M, ✓C

(8) 'One third of the cakes were eaten' [German]

a. **Ein Drittel** der Kuchen wurde aufgegessen
one third DET.GEN.PL cake.GEN.PL be.PST.3SG eaten
✓M, ✓C

b. Ein Drittel **der** **Kuchen** wurden aufgegessen
one third DET.GEN.PL cake.GEN.PL be.PST.3PL eaten
×M, ✓C

Compare all of these cases again with with American English—a language where only the count reading is available. Most important to our interests is the fact that if the complement of a fraction partitive is plural, agreement must necessarily be with that complement.

(9) One third of the walls {*is/are} moldy
×M, ✓C

To recap, the generalization can be described as the following. When a DP-external element (e.g., a verb) agrees with the complement of a FP, the interpretation is necessarily a count one. When said element agrees with the denominator (in languages where circumstances allow agreement with a denominator), there is at least a measure interpretation. Whether or not there is also a count interpretation is a point of cross-linguistic variation. The very fact that agreement is possible with the complement is mysterious. Much prior work on FPs provide a treatment where the fraction embeds the complement, yielding a DP-within-DP configuration. If we subscribe to a a theory of Agree where locality resolves competition between multiple (bundles of) features (Chomsky, 1995), only agreement with the fraction should be possible. As is clear by this point, though, this is not what we see.

The primary objective of the present study is to provide an account of this alternation in agreement and interpretation given established ideas about syntax, semantics, and their interfaces. This is done with special attention to the typological facts as I secondarily seek to explain why this alternation exists in some languages but not others. After careful examination of alternative suggestions in the literature, I ultimately propose a solution requiring an overhaul of how measurement is defined at LF; specifically, I posit that operators in the syntax can affect how variable measure functions are determined. This is a stark departure from previous work on the semantics of measure, which often assumes that the process is deterministic for pluralities and otherwise purely contextual.

The paper is structured as follows. In §2, I compare two general theories of agreement relevant to the phenomenon and provide empirical justification for one over the other. §3-4 are dedicated to the morphology and syntax of fraction partitives in the surveyed languages, providing a set of facts which should be accounted for in an analysis. We see how previous literature has derived the "canonical" count reading in §5. §6 explores two prior suggestions for deriving the measure reading that are abandoned given empirical facts. We see some crucial data from Russian that informs the present account in §7. Both the syntactic and semantic portions of the analysis are provided in §8. §9 concludes with a summary and collection of parting thoughts.

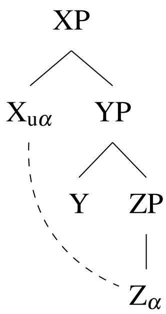
[2] Consultant found the measure reading easier to attain with a demonstrative heading the complement. The measure reading is still marginally available without a demonstrative.

2 Syntactic vs. Semantic Agreement

Before proceeding, I take this section to explore two broad ways of capturing the interaction of agreement and semantics. Deciding between these two is an important first step in developing an analysis and informs the direction of inquiry throughout this work.

The first, which I will dub SYNTACTIC AGREEMENT for sake of exposition, is characterized as the realization of a feature in multiple locations throughout a structure given some structural configuration. Some version of this conceptualization of agreement is undoubtedly familiar to syntacticians of the day, but it has been formalized many different ways throughout history. Of note is the PROBE-GOAL model of the Minimalist Program and the abundance of variations and permutations which followed. In this system, Agree is an operation between one head with an unvalued feature (probe) and another head with a feature the probe is seeking (goal), the two configured in a c-command relation.[3]

(10)



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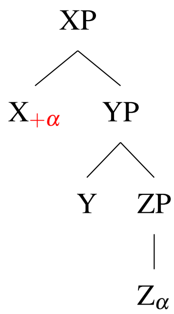
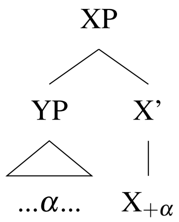


An alternative to and predecessor of the probe-goal model comes in the form of SPECIFIER-HEAD agreement (Sportiche, 1996; Koopman, 2008). As the name implies, agreement holds between a functional head and the occupant of its specifier in such a system.

(12)



These two schools of thought for syntactic agreement are distinct along several dimensions, but their bearing on the current issue is the same; deriving multiple agreement options with probe-goal or spec-head mechanics requires distinct structural representations for each agreement option. That is, the output of the Agree operation is contingent on how the target phrase is arranged (specifically, which features project to the top of the structure). Given the data presented so far, fraction partitives look the same regardless of agreement and what interpretation is licensed. Thus, one could justifiably dispute that this move is stipulative.

The competitor of syntactic agreement is the SEMANTIC AGREEMENT model (Charnavel and Sportiche, 2025a,b). Its primary innovation is proposing that the features on agreeing heads (e.g., T/Infl) are at least sometimes semantically-contentful rather than uninterpretable copies of features (contra the original formulation of Agree in Minimalism, where features copied by a probe are necessarily not interpreted). Charnavel and Sportiche entertain this possibility to account for what are often called semantic agreement phenomena,[4] characterized by an apparent mismatch between the features on a nominal and an agreeing verb. Crucially, the verb's exponed features affect how the subject is interpreted—e.g., whether *old* holds of the members or the whole in (13), whether the speaker is amongst the majority in (14). In the syntax-semantics, Charnavel and Sportiche argue that the feature on T introduces an additional presupposition about the subject (it is a plurality, it is first person, etc.) and the meaning of the overall sentence is calculated with this contribution.

[3]Chomsky's original formulation of Agree in the probe-goal model insists that the probe must c-command the goal. This is not a universal assumption amongst practitioners, though (Zeijlstra 2008 et seq.).

[4]Previous work on data like (13)-(14) suggest treatments involving what I have been calling syntactic agreement. See (Charnavel and Sportiche, 2025b, §3.1) and works cited therein.

(13) a. The committee is old [British English]
$\leadsto$ the organization was formed a long time ago $\vee$ the members of the committee are old
b. The committee are old
$\leadsto$ the members of the committee are old, $\not\leadsto$ the organization was formed a long time ago

(14) 'The majority of us are always here' [French]
a. une majorité d'entre nous {est/sont} toujours là
a majority among 1PL COP.{3SG/3PL} always here
$\leadsto$ subject may or may not include the speaker
b. une majorité d'entre nous sommes toujours là
a majority among 1PL COP.1PL always here
$\leadsto$ subject must include the speaker

At first glance, the alternation presented in §1 and the kinds of cases represented by (13)-(14) appear remarkably similar; both exhibit an interpretive alternation conditioned by morphology on an external element. Given this similarity, we might wonder if fraction partitives deserve a Charnavel and Sportiche-style account as well. While successful for prototypical cases of semantic agreement, a closer examination of the topical alternation reveals that this approach is untenable. First is the fact that the languages which exhibit variable agreement with fraction partitives do not constitute a subset of languages which exhibit semantic agreement. While French is a language with both, Italian is not (Enrico Flor, p.c.). If we were to explain the data with fraction partitives using the machinery of semantic agreement, we might wonder why it would be unavailable for alternations with group nouns, etc.

Furthermore, both measure and count readings are available for fraction partitives outside of positions of agreement—this was seen for Italian in (2), but we can observe the same for Greek in (15). Under a semantic agreement model, it is the agreeing head which licenses additional interpretations via features. This account predicts that certain readings would be impossible outside of subject position, yet this is not realized.

(15) o Giánnis éfage ená trít-o ton biskóton
the Yiannis ate.3SG one third-N.SG the cookie.GEN.N.PL
'Yiannis ate one third of the cookies' [Greek]
$^{\checkmark}$M, $^{\checkmark}$C

Finally, this model is successful with regards to grammatical number only if the alternation is conditioned by singular and plural number on the verb. However, this is not an accurate characterization of the current data. The measure interpretation is licensed by agreement with the features on the fraction, not singular; conversely, the count interpretation is licensed by agreement with the features of the complement, not plural. We know this must be the case because, in many of the languages surveyed, singular morphology on the verb becomes impossible if the numerator of the fraction exceeds two.

(16) *weil zwei Drittel der Student-en unzufrieden ist
because two third DET.GEN.PL student-GEN.PL discontent be.3SG
Int. 'Because two thirds of the students are unhappy...' [German]

(17) ??duo trit-a ton maθiton adipaθi ton Yani
two third-N.SG the student.GEN.PL dislike.3SG the Yiannis
'Two thirds of the students dislike Yiannis' [Greek]

(18) deux cinquièmes des étudiants {*a/ont} lu War & Peace
two fifth.M.PL of.DET.M.PL student.M.PL have.{3SG/3PL} read War & Peace
'Two-fifths of the students have read War & Peace' [French]

Despite its initial appeal, the above data dissuades us from pursuing a treatment of the fraction partitive alternation as semantic agreement. Instead, I opt for an analysis where the agreement facts are due to syntactic processes. By the tenet of ONE FORM/ONE MEANING, I pursue the possibility that FPs in Italian, etc. are indeed structurally ambiguous, with one structure corresponding to a measure interpretation and the other a count interpretation. It is this structural distinctness which also leads to differences in behavior in positions of agreement. In fact, there are reasons beyond agreement to motivate such an ambiguity for FPs; I elaborate upon this point in §4.

3 Anatomy of a Fraction Partitive

Let us now take a moment to more-closely examine the syntactic and morphological structure of FPs in the surveyed languages. In the process, I identify some properties of these constructions which are relevant to an analysis.

FPs in these languages can generally be decomposed into two parts: the fraction itself and the complement of the fraction. Fractions are represented mathematically with two numbers—a NUMERATOR and a DENOMINATOR—the two delimited by a fraction bar ($\frac{x}{y}$) in conventional notation. The value of the fraction is the result of numerator being divided by the denominator. Fractions in the languages covered here are largely BI-DIMENSIONAL in the sense of Anicotte (2022) in that both the numerator and the denominator are represented in the construction. This is in contrast with MONO-DIMENSIONAL fractions which refer to only one of these numbers. Russian is the only language represented in this work whose fractions (with numerator = 1) are uniformly mono-dimensional.

While the numerator slot is filled by a cardinal number or indefinite article, the denominator takes a special form of the numeral. In the languages considered, denominators outside of suppletive forms like *half* are often derived from or exactly homophonous with the number's corresponding ordinal; see Table 1. This is not a universal (see Mandarin, Japanese), but it can be observed in many non-Indo-European languages (e.g., North Sámi (Kahn and Valijärvi, 2017)). Given this generalization, it is fair to wonder whether the meaning of a fraction is derived from the meaning of an ordinal. López Palma (2011, 2012) provides an account for the semantics of fractions on this premise, but I abstract away from this detail here since it has little bearing on the puzzle at hand.

Language	**Ordinal**	**Denominator**
English	fifth	fifth
Italian	quint-	quinto
French	cinquième	cinquième
Greek	pémpt-	pémpto
German	fünfte	fünftel
Russian	pjat-	pjataja

Table 1: The number *5* in its ordinal and denominator forms by language

Representative examples of fractions in the surveyed languages can be seen in (19)-(24). Observe that in most of these languages, number and—if applicable—gender features can be seen on the denominator. In particular, the plural suffix is licensed by a numerator greater than or equal to two. This fact of fraction morphology should not come as a surprise given the nature of the puzzle and our discussion up to this point.

(19) a. one fifth
b. two fifths

(20) [Italian]
a. un terz-o
a third-M.SG
b. due terz-i
two third-M.PL

(21) [French]
a. un cinquième
a fifth.M.SG
b. deux cinquième-s
two fifth.M-PL

(22) [Greek]
a. éna trít-o
one third-N.SG
b. dúo trít-a
two third-N.PL

(23) [Russian][5]

a. (odna) vos'm-aya
one eighth-NOM.F.SG

b. sem' vos'm-ikh
seven eighth-GEN.PL

German presents an interesting counterexample as the form of the denominator stays the same regardless of the identity of the numerator. This becomes less surprising, though, when we remember that masculine and neuter nouns without syllable-final stress in German do not receive plural suffixes (Lederer, 1977, p. 171);[6] compare *Hebel* 'lever(s)' with (24).

(24) [German]

{ein/sieben} Zehntel
{one/seven} tenth

In some of these languages with a gender distinction, the gender of denominators is not homogenous; French *moitié*, Italian *meta*,[7] and German *Hälfte* 'half' are exceptionally feminine despite other denominators exhibiting masculine or neuter gender. A summary by language is provided in Table 2.

Language	**Denominator γ**
Italian	MASC, FEM
French	MASC, FEM
Greek	NEUT
German	NEUT, FEM
Russian	FEM

Table 2: Gender(s) of denominators by language

As partitive heads, fractions in these languages are to some degree privileged in having inherent gender at all; in other partitive constructions, the head will show participate in concord with the complement. Greek numeral partitives containing the cardinal *one* are a representative case.

(25) 'Thirty-one of the {men/women}' [Greek]

a. triánta énas apó tous ándres
thirty one.M.PL of the.M.PL men

b. triánta mía apó tis gunaíkes
thirty one.F.PL of the.F.PL women

These facts about number and gender morphology morphology suggest that fractions in these languages are on some level nominal items. This is bolstered by the observation that definite articles and demonstratives can appear at the left edge of FPs, given the appropriate licensing conditions. Observe that where applicable, such articles also agree in number and gender with the denominator. If covert, the numerator is inferred to be 1 in such cases.

(26) to éna trít-o
DET.M.SG one third-M.SG
'The one third' [Greek]

(27) a mozhet eta tret' byla ne toj
CONJ maybe this.F.SG third was.F.SG NEG that.F.SG
'Or maybe this third wasn't the right one?' [Russian][8]

[5] Fractions with a numerator greater than one receive genitive case (Timberlake, 2004, p. 194). As far as I know, this is separate from the paucal genitive phenomenon observed in Russian and other Slavic languages (see Pesetsky 2013 and works cited therein).

[6] I am grateful to Seva Masliukov for bringing this to my attention.

[7] I was not aware of this until Enrico Flor and Giovanni Roversi brought it to my attention, so thanks to them.

[8] From Grigory Leps - *Pannjaja tišina* (Early Silence). https://genius.com/Grigory-leps-early-silence-lyrics

Fractions can also take nominal modifiers like adjectives and relative clauses, lending further support to the previous observation.

(28) ein ungebildetes Drittel von uns
a uneducated.NOM.SG third from 1PL.ACC
'An uneducated third of us' [German]

(29) tret' studént-ov kotoraja včera xodil-a v-bar
third student-GEN.PL which.SG yesterday went-SG in-bar
'The third of the students who went to the bar yesterday' [Russian]

(30) Il terz-o del-le paret-i dipint-o da John
DET.M.SG third-M.SG of-the wall-F.PL painted-M.SG by John
'The third of the walls painted by John' [Italian]

(31) le tiers des étudiants auquel Martin a fait cours
DET.M.SG third of.DET.M.PL student-PL who.M.SG Martin have.3SG made class
'The third of the students that Martin has taught' [French]

With a clearer understanding of the fraction component of fraction partitives, let us now turn our attention to the second component—the complement. Complements in these language take the form of DPs which are introduced by a preposition or marked with genitive case. This is a language-level distinction for Italian, French, English, and Russian, but German and Greek have access to both. As far as I am able to tell, the choice of preposition or genitive case has no effect on agreement or interpretational possibilites.

(32) due terz-i del-le arance
two third-M.PL of-the orange
'Two thirds of the orange' [Italian]

(33) un sixième de la pomme
one sixth of the apple
'One sixth of the apple' [French]

(34) syem' voc'myerok jabloka
seven eighths apple.GEN
'Seven eighths of the apple' [Russian]

(35) zwei Drittel {des Apfels / vom Apfel}
two third {the.GEN apple.GEN / of.the.DAT apple.DAT}
'Two thirds of the apple' [German]

(36) to éna ékto {apó to mílo / ton mílon}
the one sixth {of the.ACC apple.ACC / the.GEN apple.GEN}
'One sixth of the apple' [Greek]

With regards to the agreement alternation on which this work focuses, we have only considered fraction partitives with plural complements. This is of course because the distinction between measure and count readings can only apply plural individuals. However, fraction partitives can also take singular DPs. In such cases, optional agreement persists for some languages, specifically French, Greek, and German.

(37) a. deux tiers de la surface sont aménagés dans un esprit lounge
two third.PL of the surface be.3PL fitted.out in a style lounge
'Two thirds of the surface is outfitted in a lounge style' [French]

b. deux tiers de la biodiversité mondiale vit dans le sol
two third.PL of the biodiversity on.earth live.3SG in the soil
'Two thirds of Earth's biodiversity lives in the soil' [French]

(38) dio trit-a tu mil-u ine kokin-{o/a}
two third-NEUT.PL DET.GEN.SG apple-GEN.SG COP.3 red-{NOM.N.SG/NOM.N.PL}
'Two thirds of the apple is red' [Greek]

(39) weil zwei Drittel einer Birne essbar {sind / ist}
because two third a.GEN.F.SG pear.GEN.SG edible {be.3PL / 3SG}

‘Because two thirds of a pear is edible’[9] [German]

Interestingly, the flexibility of agreement for plural complements does not necessarily hold with singular complements. Italian and Russian specifically forbid agreement with singular complements. American English is the reverse case, where (for some speakers) there is agreement optionality with a singular-containing FP yet not for a plural-containing one.

(40) due terz-i del-la mela { sono ross-i / *è rossa }
two third-M.PL of-DET.F.SG apple { be.3PL red-M.PL / be.3SG red-F.SG }
‘Two-thirds of the apple are red’ [Italian]

(41) dve treti étogo piroga s”edaj-em-{y/*∅} za tri minuty
two thirds this.GEN.SG cake.GEN.SG eat-PTCP-{PL/SG} in three minutes
‘One can eat two thirds of this cake in three minutes’ [Russian]

(42) Two-thirds of the eclaire was/%were already eaten by the time George found it

All of the agreement possibilities for FPs are summarized in Table 3. Providing an account for the agreement facts with singular complements is not something I strive for in this work since our focus is on the meaning alternation with plural complements. The cross-linguistic variation poses a grave challenge for any comprehensive proposal. However, I aim to provide an analysis that at least does not preclude an explanation of the singular complement facts.

	SG Comp.		**PL Comp.**	
Langs.	**Frac.**	**Comp.**	**Frac.**	**Comp.**
Am. English	%✓	✓	×	✓
German	%✓	✓	✓	✓
Italian	✓	×	✓	✓
French	✓	✓	✓	✓
Greek	✓	✓	✓	✓
Russian	✓	×	✓	✓

Table 3: Possible targets of agreement by grammatical number of complement, by language.

There are two important outcomes of this section. Firstly, fractions in these languages look a lot like nominals; they can take modifiers like adjectives and relative clauses, and they can be headed by definite items like demonstratives or relative clauses. Furthermore, they come with their own number and (if applicable) gender features. The moral of the story is that a successful analysis will treat fractions as such, and—in conjunction with the conclusion of §2—features on agreeing verbs are necessarily copied from the fraction itself. This seems trivially true, but it poses a challenge to alternatives which are explored in §6. Secondly, FPs can take singular complements and (in some languages) still exhibit the agreement optionality introduced previously. In developing a semantic for fractions, we must ensure that they can handle both singular and plural individuals. Furthermore, the analysis should not impede agreement optionality for singular-containing FPs in the languages that allow it.

4 Syntactic Correlates of Agreement

Recall that in §2 I present a series of arguments for the measure and count readings corresponding to distinct structures rather than the same. Here, auxiliary evidence for this position is provided in the form of syntactic diagnostics. This is also done for the sake of informing an analysis of measure and count FPs.

I showed previously that fractions in the surveyed languages can take nominal-domain modifiers like demonstratives, determiners, relative clauses, and adjectives. What I neglected to mention there, though, is that the presence of such elements precludes DP-external agreement with the fraction in many cases.[10]

[9] Note that some speakers of German cannot have Agreement with denominator if the complement is definite.

[10] Bizarrely, American English represents an exception by maintaining complement agreement even with a restrictive relative clause modifying the fraction.

(43) ein ungebildetes Drittel von uns {hat/*haben} für ihn gestimmt
one uneducated.NOM.SG third from us have.{3SG/1PL} for him voted
'An uneducated third of us voted for him.' [German]

(44) krasnaja tret' stulje-v stojal-{a/*i} v-uglu
red.NOM.F.SG third chair-GEN.PL stand-{SG/PL} in-corner
'The red third of the chairs are standing in the corner' [Russian]

(45) Il [terz-o del-le paret-i dipint-o da John] {è ross-o / *sono ross-e}
DET.M.SG [third-M.SG of-the wall-F.PL painted-M.SG by John] {be.3SG red.M.SG / be.3PL red-M.PL}
'The third of the walls painted by John are red' [Italian]

(46) 'The third of the students that Martin has taught have studied semantics' [French]

a. le [tiers des étudiants auquel Martin a fait cours] a étudié la sémantique
the [third of.the student-PL who.M.SG Martin have.3SG made class] have.3SG studied the semantics

b. *le [tiers des étudiants auxquels Martin a fait cours] ont étudié la sémantique
the [third of.the student-PL who.M.PL Martin have.3SG made class] have.3PL studied the semantics

In at least Russian, German, and Greek, definite elements (determiners, demonstratives) at the left edge of the FP also blocks agreement with the complement.

(47) èta tret' studént-ov sobral-{as'/*is'}
this.F.NOM.SG third student-GEN.PL gather-{SG/PL}
'This third of the students gathered' [Russian]

(48) das Drittel der Wände wurd-{e/*en} gestrichen
DET.NOM.SG third DET.GEN.PL wall.PL be.PST-{3SG/3PL} painted
'The blue third of the walls were painted' [German]

(49) to ena trito ton milon ine {kokino/??kokina}
DET.N.SG one third DET.GEN.PL apple.GEN.PL COP red.{N.SG/PL}
'The one third of the apples are red' [Greek]

PP-extraposition precludes agreement with the complement in Greek and German.

(50) ena trít-o ___ ine kokin-{o/*a} apo ta mila
one third-M.SG COP.3 red-{M.SG/PL} of the apple.PL
'One third of the apples are red' [Greek]

(51) ein Drittel ___ wurde-{∅/*n} gespült von den Tellern
a third be.PST-{3SG/PL} washed of the.DAT plate.DAT.PL
'A third of the plates were washed' [German]

French and Italian—languages with partitive-internal cliticization processes (Boivin, 2005)—can only cliticize the complement if agreement is with the fraction.

(52) Un tiers en {a été mangé / *ont été mangées}
a third of.them {be.3SG been eaten.M.SG / be.3PL been eaten.F.PL}
'A third of them (the apples) have been eaten' [French]

(53) Un terzo ne {è stato mangiato / ??sono state mangiate}
a third of.them {be.3SG been.M.SG eaten.M.SG / be.3PL been.F.PL eaten.F.PL}
'A third of them (the apples) have been eaten' [Italian]

The evidence put forth in this section bolsters the argument I made in §2; despite superficial similarities, measure

(i) The [one third of the students who like John] also **like** Mary.

and count FPs actually correspond to separate structures. The facts observed here will ultimately inform the shape of the measure and count structures in §8.

5 How to Derive Count

Let us now take a moment to review how the more-conventional measure reading is derived in a formal semantic system per the literature. Descriptively, there are two components to an FP's semantics. First is a condition of parthood; the referent of a fraction partitive must be a proper part of the individual denoted by the complement.[11] Secondly, there is a comparison of degrees between the referent and the complement. The measure of the individual denoted by the FP on some scale is the measure of the complement on the same scale divided by the denominator. Intuitively, if a rock is three kilos, a third of that rock will be one kilo.

As seen in §4, FPs can take either singular or plural individuals as their complement. This is a feature of many partitive constructions in many of the world's languages (Moltmann, 1997; Wagiel, 2021), extending even to languages like Passamaquoddy (Algonquian), which is very much unlike the European languages we've been considering up to this point.

(54) a. epahsiw cikon wekine
half apple.PROX.SG spoil.AI
'half of the apple spoiled'
b. epahsiw cikoni-yik wekini-yik
half apple-PROX.PL spoil.AI-PROX.PL
'half of the apples spoiled' [Passamaquoddy]

Thus, a formal definition for FPs presupposes a shared type between singular and plural DPs. I follow Link (1983) et seq. in treating pluralities as individuals (type e) rather than as sets ($\langle e,t\rangle$, cf. Bennett 1974; van der Does 1991; Winter 2001). Borrowing heavily from Classical Extensional Mereology (Lesniewski, 1931; Leonard and Goodman, 1940), he treats pluralities as mereological sums of "atomic" individuals (singularities), which are of course also type e. This nets the selectional flexibility for FPs that we seek, but there is a lingering problem. In this system, membership of a plurality is modeled with the mereological relation (PROPER) PART-OF ($\sqsubset$).[12] However, Link distinguishes two kinds of parthood which relate different kinds of objects. One, ATOMIC *part-of*, holds between pluralities and the (sums of) individuals which compose them. The other, MATERIAL *part-of*, holds between singular individuals and their parts. This is done in service of the observation that predicates can apply to singularities but not necessarily the sum of their parts; as the famous example goes, a new ring can be made of old Egyptian gold. So, while both pluralities and singularities are type e, following Link's system would predict that FPs—which presuppose parthood—could only be compatible with one or the other, else fractions are ambiguous between a singularty-selecting form and a plurality-selecting form.

This flies in the face of Moltmann and Wagiel's observation about partitive constructions, prompting them to abandon the distinction between atomic and material part-of. With regards to the denotation of partitive constructions, though, this analytical move creates a separate problem. Even the simplest mereologies assume that parthood has the property of being transitive (Varzi, 2003) (55). If we maintain this for natural language, we would have to accept that the referent of a partitive containing a plurality could be a material part of an member of the plurality. For example, let's say there is a plurality of New Yorkers of which Jerry Seinfeld is a member. Of course, Seinfeld's right hand is a part of him, and he is part of the New Yorkers by virtue of being a member of that plurality. This would predict that Seinfeld's right hand is in the extension of *part of the New Yorkers*, yet Seinfeld's hand and nothing else being completely covered in paint is clearly not a verifying condition for (56). Let's call this the TRANSITIVTY PROBLEM.

(55) *Transitivity*
$\forall x,y,z.(x \sqsubseteq y \wedge y \sqsubseteq z) \rightarrow x \sqsubseteq z$
(Part of part of a whole is itself part of that whole.)

[11] Some works (e.g., Ionin et al. 2006; Magri 2008) present an analysis of partitives s.t. *of* and its equivalents are the locus of the parthood assertion. While this is a plausible analysis, I do not follow this in the present work because—as far as I am able to tell—it confers no analytical advantage and defining fractions with parthood baked-in is notationally simpler.

[12] In a system that treats pluralities as sets, this would naturally be captured with a set-membership or (proper) subset relation.

(56) Part of the New Yorkers are completely covered in paint

Moltmann and Wagiel seek to restrain the meaning of plural-containing partitives to pick out only singularities and their sums, excluding the material parts. To avoid the apparent overgeneration, they posit that the transitivity of parthood can fail under certain circumstances. Specifically, for any given pair individuals, one can only be part of another if there is no intervening individual which is an INTEGRATED WHOLE (57). This is what Moltmann and Wagiel call the TRANSITIVITY PRINCIPLE and TOPOLOGY-SENSITIVE TRANSITIVITY, respectively.[13]

(57) $\forall x,z.x \sqsubset z \rightarrow \neg\exists y.x \sqsubset y \wedge y \sqsubset z \wedge \texttt{int-wh}(y)$

How an integrated whole is defined varies between the two authors. Moltmann appeals to R-INTEGRATED WHOLES (Simons, 1987), which are defined in (58). To Wagiel, an IW is an individual that has the mereotopological property of being MAXIMALLY-STRONG SELF-CONNECTED (Casati and Varzi, 1999) relative to some property. Essentially, an MSSC individual is one who is SELF-CONNECTED (any two ways of dividing an individual are connected), has an interior which is self-connected, and the self-connected things which overlap with it are part of it. A formal definition is given in (61). Regardless, the end result is the same—a materially-real, single, whole individual like Jerry Seinfeld constitutes an IW whose parts are not members of any plurality said IW is part of. I dub this strategy for resolving the transitivity problem FORMALIZED TRANSITIVITY SUSPENSION (FTS). The approach and its efficacy will be revisited in §8.1.

(58) An entity x is an R-integrated whole if there is a division of x such that every member of that division stands in the relation R to every other member and no member bears R to anything other than members of the division. (Moltmann, 1997, 24)

(59) Self-connected
$\texttt{SC}(x) \stackrel{\text{def}}{=} \forall y \forall z[\forall w(\texttt{overlap}(w,x) \leftrightarrow (\texttt{overlap}(w,z))) \rightarrow \texttt{connected}(y,z)]$
(x is self-connected iff any two parts that form the whole of x are connected to each other.)

(60) Strongly self-connected
$\texttt{SSC}(x) \stackrel{\text{def}}{=} \texttt{SC}(x) \wedge \texttt{SC}(\texttt{interior}(x))$
(x is strongly self-connected if x is self-connected and the interior of x is self-connected.)

(61) Maximally-strongly self-connected
$\texttt{MSSC}(x) \stackrel{\text{def}}{=} \texttt{SSC}(x) \wedge \forall y[\texttt{SSC}(y) \wedge \texttt{overlap}(y,x) \rightarrow y \sqsubseteq x]$

With parthood out of the way, let us consider how to model the degree comparison portion of fractions. I assume an ontology which includes measure functions that take individuals and return degrees ($\langle e,d \rangle$). Crucially for our purposes, arithmetical operations can apply to degrees (i.e., they can be added, multiplied, divided, and subtracted). Forgetting momentarily about partitives with plural complements, the measure function of a fraction must be somewhat flexibile; compare *a third of the cake*, where the relevant dimension of comparison is likely volume, and *a third of the movie*, where the dimension is likely runtime. To account for this, Wagiel builds on work by Bale and Barner (2009), introducing a variable measure function μ. The output of this measure function, he proposes, is partially situational—given the dimensions along which a certain individual is measure, what measure function actually surfaces is defined by the context. By further assuming that a cardinality function which counts the integrated wholes that compose an individual is another possible realization of μ, both mass partitives and set partitives are generable with (62).

(62) $[[\text{third}]] = \lambda y_e \lambda x_e . x \sqsubset y \wedge \mu(x) = \mu(y) \times \frac{1}{3}$

How do we get specifically the count reading for FPs? As it stands now, (62) could pick out a part of a plural individual by mass, volume, etc. if context allowed for it. To guarantee that only a count reading emerges, he imposes a partial order on the set of (non-monotonic) measure functions where, importantly, cardinality precedes every other kind of measure. This set, which I'll call M, is shown in (63). By further stipulating that the most minimal, possible measure function will be chosen for μ, Wagiel has guaranteed that μ will count an individual as long as it is countable (i.e., is a plurality).[14]

[13] Moltmann's view on what a part can be goes beyond the transitivity principle. She also notes that in certain cases it can depend on the whole itself what the parts are, which is explored in Moltmann (1998, prep).

[14] Wellwood 2019, Ch. 5 similarly posits that a variable measure function which takes a plurality must be a cardinality function,

(63) $M = \{\mu_{card} < \mu_{vol}, \mu_{runtime}, \ldots\}$

(64) Contextual conditioning (Wagiel 2021, citing Bale and Barner 2009)
μ is interpreted as one of the measure functions m_z in the series $\langle m_1, m_2, m_3, \ldots m_n \rangle$ such that the argument for μ is in the range of m_z; furthermore, contextually m_z is preferred to m_y if $m_z < m_y$

To sum up this section, we motivated a semantics for fraction partitives which is flexible enough to deal with both singularities and pluralities. Tools from mereology, plus a constraint on parts of pluralities (i.e., sums of integrated wholes) characterize the parthood component. Contextually-defined measure functions which are additionally biased towards counting for pluralities handles the degree comparison component. Specifically, the count reading follows because of the ordering of the set in (63). Of course, these are insufficient for capturing the measure reading we observe in Italian, etc. In the following sections, I explore our options for supplementing or modifying a Wagiel-like account to derive the measure reading.

6 Previous Attempts at Deriving Measure

Despite the ubiquity of the count reading for FPs in the literature, the measure reading has not gone completely unnoticed. Something like it has been suggested at least since Hoeksema (1996) who argues that a FP like *half of Jane and Jacky* most naturally refers to the sum of half of Jane and half of Jacky rather than one of them individually.[15] However, the fact that in some languages the measure reading is licensed only by a certain agreement configuration is an original observation; as far as I am aware at least, no author up to this point has made the connection.

In this section, I present two prior suggestions on deriving the measure reading for FPs. Where possible, I make a good faith attempt to extend the analysis in order to capture the agreement facts. Ultimately, both options are abandoned for their incompatibility with the data (some of it previously discussed, some of it novel).

6.1 The de Hoop Maneuver

In her account of the Partitive Constraint, de Hoop (1997) proposes that (most) plural DPs are ambiguous between individuals (e) and sets ($\langle e,t \rangle$). She further posits that some partitives differ in what kind of semantic type they select for; most relevant to us, the heads of proportional partitives (fractions, but also percentages, etc.) can take either type as an argument. Thus, the difference in meaning is resolvable to the semantic type of the complement. Under her account, there are no restrictions on what part of a plurality can denote (cf. the discussion in §5), whereas subsets must be comprised of individuals in the relevant set. In de Hoop (2003), she writes:

> I assume that Jane and Jacky can denote a complex entity, which explains the grammaticality of *half of Jane and Jacky*[.] At the same time, I conclude that *Jane and Jacky* cannot denote a set of entities and this accounts for the ungrammaticality of **one of Jane and Jacky*[.] Note that *half of Jane and Jacky* does not refer to Jane or Jacky. If we consider *Jane and Jacky* to denote a composed entity, then half of it can be any half.

Given what she has said up to this point, a fraction must be able to take sets and individuals. To handle this, we could plausibly posit higher types (something which is neutral between e and $\langle e,t \rangle$) and posit a relation to stand in for both *part-of* and *subset-of* (Lewis, 1991). She does not say anything about how the relevant measure function would be defined, but I conjecture there could be a rule that makes cardinality the only viable measure for a set argument.

although she appeals to slightly different formal machinery.

[15] You will notice here and in the remainder of the section that these explorations of the interpretation of FPs cite English data. I personally disagree with the many of their judgments for my own English and maintain that the measure reading for FPs is not robustly attested in American English. For cases involving conjunction, the preferred reading is very plausibly a case of conjunction reduction (*half of Jane and ~~half of~~ Jacky*) and thus distinct from the measure reading we've discussed so far. Falco and Zamparelli (2019) (discussed at length in §6.2) report similar readings for FPs containing definite plural DPs, but as far as I am able to tell none of them are native American English speakers. I have shown that the measure reading is observed in many of the languages spoken on Europe including Italian, so I suspect judgments from their respective L1s are influencing how they can interpret these FPs. I discuss the works regardless because their descriptions of the phenomena match that of measure readings and their proposed solutions are plausible approaches to analyzing measure FPs.

If we assume that de Hoop is correct about proper parts of plural individuals (i.e., there is no transitivity suspension), her account gets us both readings. A number of issues can be identified, however. Firstly, this analysis does not lend itself to a clean account of the cross-linguistic variation. If American English does not have a measure reading, de Hoop would have to say that the relevant plural DPs can only denote sets in this language; can de Hoop commit to this position? Falco and Zamparelli (2019) (discussed further in the following subsection) present a series of arguments against her position. For one, treating definite plurals as $\langle e,t \rangle$ opens them up to semantic processes typically reserved for NPs and predicates (quantification, predicate modification, etc.), making forms like **red the flowers* possible without further stipulation. Furthermore, what has the privilege of $e \sim \langle e,t \rangle$ ambiguity in her system is ill-defined. She argues that coordinated singulars can only be e, yet a coordinated DP like *the North Pole and the South Pole* picks out the same plurality as *the Earth's poles*, a definite plural—which can be either e or $\langle e,t \rangle$ according to de Hoop.

An attempt to extend this proposal to account for agreement facts is largely unsuccessful. Intuitively, one way forward would be to say that the kind of agreement you get is contingent on the semantic type of the complement, ignoring the features on fractions. However, there is nothing in de Hoop's theory that says that type ambiguity is restricted only to the complements of partitives. She implicitly assumes that this type ambiguity is actually all over language, yet only detectable in certain syntactic-semantic contexts (i.e., in partitives). If we take this path to explain agreement, we would now expect agreement optionality to show up anytime you're agreeing with a plural DP. While it's true that there are some other agreement alternations in these languages (e.g., semantic agreement, default agreement), we do not observe the wholesale optionality that this line of reasoning would have us expect. Additionally, this boxes us into the assumption that the measure reading is licensed by a singular feature on the verb rather than the verb copying the features of the fraction. This is problematic for all the reasons mentioned in my discussion of semantic agreement in §2.

6.2 The Falco & Zamparelli Gambit

In explaining the measure reading for FPs, Falco and Zamparelli (2019) end up taking an entirely different path for deriving the measure reading. Their suggestion coincidentally pairs well with Wagiel's method for deriving count FPs. Recall that his account is contingent on two things: transitivity suspension (proper parts of an individual x are not parts of the plurality which x is a part of) and how variable measure functions are defined (μ will be a cardinality function if it takes a plural argument, per Wagiel and Bale and Barner (2009); Wellwood (2019)). If we want to hold these two aspects of an analysis of FPs constant, one play might be to "trick" the fraction into thinking its complement isn't really a plural individual.

Falco & Zamparelli take exactly this approach. Specifically, they suggest that the measure reading is the result of a covert operator which takes the plural complement and returns unindividuated mass—perhaps the UNIVERSAL GRINDER of Pelletier (1975); Pelletier and Schubert (1989). We can define a UG as in (65), which takes an individual x and returns a new individual which is the sum the set of zs s.t. z is a part of y and y is a proper part of x.

(65) $[\![\text{UG}]\!] = \lambda x. \bigoplus \{z : \forall y. z \sqsubseteq y \wedge y \sqsubset x\}$

Given transitivity suspension, this operator will return an object which is distinct from the original individual. For starters, while Jerry Seinfeld's hand is not a part of the linguists, his hand is a part of $[\![\text{UG}]\!]([\![\text{the linguists}]\!])$. Furthermore, if the means of individuating integrated wholes has been destroyed, it will now be impossible to count the members of pluralities. Thus, to Falco & Zamparelli, the measure reading of the fraction partitive will look something like this:

(66) [terzo [UG [$_{\text{DP}}$ le pareti]]]

How might we extend the proposal to capture Agreement facts? Under a system like Deal (2015)'s INTERACTION & SATISFACTION, it is possible to posit probes which will select for a less-local goal over a more-local one if the former has some set of features which are more "desirable". Given that the measure reading is licensed by agreement with the fraction, we could imagine that some aspect of the complement's features spur a probe to Agree with the fraction instead. Is it possible that the mass-ishness of an individual taken by the UG is responsible?

Unfortunately, this venture doesn't get us very far. If the analysis I sketched above is correct, we would expect it to be impossible for a verb to agree with the complement of any FP if that FP were a canonical mass noun. Data from German and Greek shows us that this prediction is not realized, suggesting that if this is indeed an effect of a Universal Grinder, fraction agreement proceeds a different way.

(67) zwei Drittel des Wassers {ist / sind} vergiftet
two thirds DET.GEN.SG.M water.GEN {COP.3SG / COP.PL} poisoned
'Two thirds of the water is poisoned' [German]

(68) duo trita ton çumos ine pradin-{a/os}
two third.N.PL DET.GEN.M.SG juice.GEN COP green-{N.PL/M.SG}
'Two thirds of the juice is green' [Greek]

Putting the issue of implementing the agreement facts aside, there is a more serious problem with their suggestion suggestion. Many languages with a grammatical number distinction on nominals allow mass nouns to be "bare"—i.e., without non-singular number morphology or articles.[16] The same is true for count nouns which have been coerced into having a mass interpretation; in French and English, for example, canonically-count nouns appear bare under a mass interpretation (69)-(70). This is straightforwardly explainable if we assume count-to-mass operations apply to NPs (sets of atoms/IWs) instead of DPs (71) (Rothstein 2009; a.o.). Such an operation, I conjecture, would block the application of plural morphology.

(69) Il y a assez de **tapis** pour nous quatre
EXPL there have.3SG enough of carpet for 1PL four
'There is enough rug for the four of us' [French][17]

(70) **The three squirrels** carelessly wandered into traffic and were struck by a motorist.
Now there is **squirrel** all over the road.

(71) $[[\text{UG}]] = \lambda P_{\langle e,t\rangle} \lambda x_e.\exists y.y \in \pi_i(P) \wedge x \sqsubset y$ Adapted from Kiss et al. (2021)

However, under a measure interpretation of an FP—which F&Z would analyze as the result of a UG applying to the complement—we observe no change on the morphology of the complement in the languages which exhibit this alternation. It is not impossible that the operator we would need to capture the measure reading is distinct from that which is responsible for more-familiar count-mass conversions. This would predict, though, that mass interpretations of plural DPs could be found elsewhere in these languages. At present, I am aware of no such evidence.

The gravest problem to this account comes from semantics, though. If the measure reading is really resolvable to the presence of a universal grinder, we should at least expect there to be limits to what kinds of predicates a measure FP can combine with. The referent of a measure FP should also denote a mass by virtue of having a mass DP as its complement. At minimum, they should be incompatible with distributive/individual level predicates—observe a straightforward example in (72).

(72) An army of Boston Dynamics robots were smashed to pieces by the anti-AI protesters.
#Now robot is (each) begging for mercy.

What we observe, though, is that FPs are compatible with these predicates even with fraction agreement (which, recall, licenses the measure interpretation). Since these predicates range over pluralities, I conclude that measure FPs can (but do not necessarily) denote pluralities. This is a fact that Falco & Zamparelli's approach cannot explain because, to them, the emergence of the measure reading is based entirely upon transforming a plurality into unindividuated mass. The application of a grinder to the complement of the FP predicts downstream effects which are not observed.

(73) to ena trito ton vivlion zugiz-ei 4kg to kaθena
DET.NOM.SG one third DET.GEN.PL book.GEN.PL weigh-SG 4kg DET each.N
'One third of the books weigh 4kg each' [Greek]

(74) un cinquième des garçons {a/ont} bougé un piano
a fifth of.DET.PL boy.PL have.{3SG/3PL} moved a piano

[16] Recent research has shown that the mass nouns of some languages morphologically pattern with plurals over singulars in some respects, though. Nukuoro (Austronesian > Polynesian) is a representative case (Asperheim and Rudolph, 2025).
[17] Adapted from (Bosque, 2024, p. 139).

‘A fifth of the boys moved a piano’ (✓DISTRIBUTIVE) [French]

(75) ena trit-o ton maθit-on ine arost-{o/i}
one third-NEUT.SG DET.GEN.PL student-GEN.PL COP.3 sick-NOM.N.SG/-NOM.M.PL
‘One-third of the students are sick’ [Greek]

(76) Un terzo del-le ragazz-e {é andat-o / sono andat-i} in spiaggia
One third of-DET.F.PL girl-F.PL {have.3SG walked-M.SG / have.3PL walked-F.PL} to beach
‘One third of the girls walked to the beach’ [Italian]

6.3 Summary

Let us briefly take stock. In this section, I reviewed two previous accounts of the semantic alternation observed in FPs. I also made, where possible, attempts to reimagine these proposals to account for the agreement facts. While very different in a number of ways, both place the burden of the interpretational change on the complement of the FP. For de Hoop, it was a matter of type ambiguity; for Falco & Zamparelli, a covert operator applied to the complement and changed its properties to get the measure reading. The latter is particularly enticing not only for its intuitiveness, but also for its tight compatibility with the assumptions we need to derive the count reading in §5. As I have hopefully shown, however, these angles are unsatisfying at best and potentially quite problematic at worst. This is all without mentioning that they would have no account for the syntactic alternations found in §4 since they are operations which apply only to the complement. Qualities of the embedded nominal make no conceivable predictions about, e.g., whether an adjective or demonstrative could be present at the left edge of a FP. I conclude based on available evidence that we should not place the onus of interpretational difference on the complement.

What, then, is the alternative? If the difference between measure and count is not an effect of the complement, perhaps it is the fraction (or something in the “fraction-field” of the FP) which licenses an alternative denotation. In the following sections, I explore just such a possibility with a particular piece of data from Russian.

7 Insights from Russian *čast’*

In addition to the “simple” fraction partitives we have seen up to this point, Russian additionally has a FP construction which overtly includes the part-word *čast’*. Its relevance to the current puzzle is two-fold. Firstly, agreeing verbs can only show feminine agreement when *čast’* is present; complement agreement is suddenly impossible. Additionally, only the measure reading is accessible in this configuration.

(77) tret’ja **čast’** sten krasn-{✓aja/*ye}
third.ADJ.F.SG part.F.SG wall.F.PL red-{F.SG/PL}
‘A third part of the walls are red’ [Russian]
✓M, ×C

Also note that when the numerator is greater than one, *čast’* is marked genitive plural (*časti*). This case-marking is a characteristic of Russian bare FPs as well (see §3, *fn.* 5).

(78) sem’ vos’myx časti pirog-ov
seven eighth.GEN.PL part.GEN.PL cake-GEN.PL
‘Seven eighths of the cakes’ [Russian]

Russian FPs containing *čast’* bear an obvious resemblance to the measure FPs I’ve discussed in this work (see Table 4). It stands to reason that, given this striking similarity, the two are perhaps derived from the same underlying structure. In the remainder of this paper, I pursue an analysis involving a covert operator which shifts the denotation of a FP from count to measure; I take *čast’* to be the optionally-overt realization of this operator.

This data and the way I interpret it should be qualified before proceeding. Despite appearances (and how it is glossed), I do not take this use of *čast’* to mean *part* in the more-familiar sense. Both measure and count FPs require that their referents be proper parts of their respective complements, so treating this item as such would be redundant. Furthermore, some languages have superficially similar FPs (i.e., ones which include an overt *part*-word) yet they exhibit completely different behavior. German fractions are commmonly analyzed as

	M	C
<u>tret’</u> sten	✓	✓
tret’ <u>sten</u>	×	✓
<u>tret’ya</u> <u>čast’</u> sten	✓	×

Table 4: Interpretation possibilities by FP type (<u>underline</u> = agreement)

a phonological reduction of an ordinal (*dritte*, *fünfte*, etc.) plus the *part*-word *Teil* (Lederer, 1977). Yet as we saw, there is optionality in terms of agreement and interpretation.

English also has (or at one point had) such a construction as well, exemplified in (79). It is difficult to comprehensively evaluate the semantic import of *part* in these FPs because they seem quite marginal in contemporary Englishes; as of 2025, the Corpus of Contemporary American English (CoCA, Davies 2010) contains 7613 instances of *a third of* but only 23 instances of *a third part of*. Similarly, the British National Corpus (BNC, Davies 2004) contains 949 instances of *a third of* but a mere 3 instances of *a third part of*. Semantics aside, (79) shows that at minimum the presence of *part* does not preclude agreement with a complement, unlike in Russian.

(79) [A] third part of the waters **were** turned into wormwood (CoCA, Davies 2010)

Another point of difference between Russian and English FP-*part* constructions is that while *čast’* shows plural morphology when the numerator of the fraction is greater than one, English data from the web would suggest that this is not a requirement for *part*.

(80) the two thirds parts of my personal estate that I have in Kings County or elsewhere[18]

(81) He had to pay two-thirds part of a piece of silver to have the plows and picks sharpened[...][19]

The details of English FP-*part* constructions are interesting and warrant further investigation. Chief among these questions is what the semantic difference between this and bare FPs might be, if any. However, it is beyond the scope of the present work. The agreement and plural concord facts—taken in conjunction with the observation about German fractions—are enough to dissuade us from an analysis where the measure reading arises from the addition of *part*-words. This scores us some flexibility in how we can define the semantic function of *čast’* and the hypothesized covert equivalent.

8 Proposal

This section details a unified analysis for the topical alternation inspired by the facts about Russian FPs shown in §7. In short, I reduce the measure/count contrast to the presence or absence of a special operator which I call a MEASUREMENT SHIFTER (M-SHIFT). I reason that it can be optionally overt in languages like Russian, where it is realized as *čast’*. I will assume the Y-model of the grammar, where syntax builds structure via Merge and Agree before shipping the structure to interpretational and phonological components. M-SHIFT is merged with the fraction partitive, manipulating its syntax to make the fraction accessible to other syntactic operations (most importantly, Agree). When this structured is sent to LF, M-SHIFT adjusts how measure functions are defined so that pluralities cannot be counted. Without M-SHIFT, Agree will target the complement of the FP and the count reading will emerge roughly as outlined in §5.

Treating the measure reading as the more-marked case is supported by the positive evidence we see in Russian, but it has the added benefit of explaining the American English data. Recall that in this language, only the count reading is accessible and fraction agreement is impossible with plural complements. Making the measure reading and its corresponding agreement a function of M-SHIFT resolves the variation to the lexicon—Italian, Greek, and so on have M-SHIFT, whereas American English does not. Something else will be have to be said about German and Russian, though, since the count reading is not blocked with fraction agreement.

[18] From the last will and testament of John Van Wickele, 1731.
https://facultysites.houghton.edu/JohnVanWicklin/Home%20page/Genealogy/FamPages/jan%5E3will.htm

[19] 1 Samuel 13:21, NLV. https://www.biblegateway.com/verse/en/1%20Samuel%2013%3A21

8.1 At LF

Let us first examine the semantic function of M-SHIFT and how it yields the measure reading. To get this analysis off the ground, we need to make two changes to the analysis in §5. To begin with, I maintain Wagiel and Moltmann's position that a single parthood relation holds between integrated wholes, parts of integrated wholes, and pluralities. However, I deviate from them by uncodifying formalized transitivity suspension and insisting that parts of integrated wholes be parts of pluralities. That is, while a material part of a single wall or sums of material parts of multiple walls would not be in the extension of (82) for Wagiel and Moltmann, it is for me. This is necessary because (as discussed in §1) measure FPs can denote things outside integrated wholes and their sums; (2) could be true in a situation where a third of every wall is covered in mold, or one where a single region of one wall is covered in mold, etc.

(82) $\lambda x.x \sqsubset [[\text{the walls}]]$

However, the transitivity problem which motivated FTS still remains. Without it, though we have no explanation for the badness of (83) and similar cases. How can we account for this?

(83) ??/#Cooper's hand is (a) part of the linguists

What I provisionally suggest is that the denotation of plural-containing partitives is a product of pragmatics rather than semantics. This can be illustrated with what I call the PARTHOOD-SUBSET MAXIM. Given formally in (84), it states that the set of a's parts will always be a proper subset of b's parts as long as a is a proper part of b. If assume a gunky universe where every entity has a proper part and thus there are no atoms in the traditional sense, both sets will have a cardinality of ∞. Yet, a's parts will constitute a proper subset of b's parts because b contains a (and thus a's parts are included in the set of b's parts).

(84) Parthood-Subset Maxim
$\forall a, b.(a \sqsubset b) \rightarrow (\{x : x \sqsubset a\} \subset \{y : y \sqsubset b\})$

If we ignore transitivity suspension, this will also hold for singularities and the pluralities of which they are members. Given a plurality like *the New Yorkers*, the set of its parts will be a superset of, e.g., the set of Jerry Seinfeld's parts because the parts of non-Jerry New Yorkers are also included in the set of New Yorker parts. Ergo, a partitive containing a plurality is far vaguer than one containing a singularity which is a member of said plurality. Conversely, the latter is far more informative than the former. Compare this fact with canonical cases of scalar implicature involving quantifiers, where on literal interpretation the extension of a declarative[20] containing a universal quantifier is a proper subset of the extension of one containing an existential quantifier.

(85) $\{w : \text{George ate } \forall \text{ eclaires in } w\} \subset \{w' : \text{George ate } \exists \text{ eclaires in } w'\}$

Conversational reasoning à la Grice (1975) states that a cooperative speaker will choose the most informative thing that an epistemic state allows them to commit to (the MAXIM OF QUANTITY). As has been discussed by Grice and the vast body of literature which followed his proposal, this means that a speaker will use an existential quantifier when e.g., some but not all of the eclaires were eaten despite the fact that an existential quantifier is logically compatible with a universal one. With regards to partitives, a cooperator speaker should use partitive DP.SG when they want to talk about a subatomic part of a singularity (even though partitive DP.PL literally contains the meaning of partitive DP.SG). In turn, if a listener hears partitive DP.PL, they can reason that the speaker did not mean partitive DP.SG, strengthening the meaning of the plural to an exclusive interpretation. In appealing to strengthening via pragmatics, we can account for the transitivity problem while preserving a simple semantics for parthood (i.e., one which is lifted directly from Lesniewski (1931); Leonard and Goodman (1940)).

However, this account predicts that the implicature I outlined above can be obviated in cases of epistemic uncertainty, thus readmitting the "inclusive" reading of a plural partitive. Such effects have long been observed for plurals in English and a variety of other languages (Sauerland, 2003; Sauerland et al., 2005). These items are conventionally understood to denote individuals whose cardinality is greater than one, but they can in fact denote singularities in questions or under universal quantifiers, for example. If these partitives are like plurals or

[20] Assuming that a declarative sentence denotes worlds in which that statement is true.

existential quantifiers in having a weak meaning which is strengthened, we would expect them to be able to pick out subatomic parts in these same environments.

Preliminary evidence from English suggests that this prediction is substantiated. Suppose the Seinfeld gang has been tasked with repainting a fleet of boats. In (86), the answer to the question is positive even if George only painted part of a single boat. Similarly, patients of mixed sizes are possible under universal quantification, as opposed to only integrated wholes and their sums (87). Furthermore, in cases where it is unclear which member of a plurality a subatomic part came from, the predicative use of a partitive DP.PL suddenly does not seem so bad (88).

(86) Q: Has George painted part of the boats?
A: Yes; in fact, he painted the starboard side of USS Anchorage.

(87) Every New Yorker painted part of the boats. Elaine painted part of USS Bataan, Kramer painted USS Anchorage and USS Ticonderoga, Jerry painted USS Arlington, and George...

(88) This propeller must be (a) part of the boats (but I'm not sure which one).

Also like plurals, homogeneity-like effects can be observed under negation. Given the analysis of negative quantifier *no* in (89), the contradiction in (90) should not appear if a sub-ship part is not in the set denoted by the partitive. The fact that it does is not straightforwardly explainable under FTS, lending further evidence to the proposed reanalysis.

(89) $[[\text{no}]] = \lambda P_{\langle e,t\rangle} \lambda Q_{\langle e,t\rangle}.\neg\exists x.P(x) \wedge Q(x)$

(90) No part of the boats {was/were} painted. #Jerry (only) painted half of USS Ticonderoga.

Our interim conclusion is that plural partitives can in fact denote part of a singuliarity in said plurality. It is the pragmatics which generates the conventional meaning, and this can be precluded under the right conditions.[21] The sum of these observations persuade us that FTS is plausibly unncessary for an account of parthood and partitivity. Ergo, fraction partitives can refer to parts of the plurality which are not integrated wholes or sums of integrated wholes (without deferring to something like a universal grinder as Falco & Zamparelli did).

The second change I make to the assumptions laid out in §5 concerns the set of measure functions M. Recall that this set, whose members are the non-monotonic measure functions that a variable measure function μ can be defined as, is partially-ordered. Specifically, it is ordered so that cardinality is outranked by all other forms of measure. Wagiel adapts an algorithm from (Bale and Barner, 2009), paraphrased as follows: given an individual x, μ is defined as a measure function in M which (i) ranges over x and (ii) is minimal with respect to the ordering of M. If multiple measure functions satisfy these two properties, context decides how μ is realized. In Bale & Barner's formulation of this set, only one element is minimal because there is only one ordering relation (cardinality < everything else). This makes it so that when a variable measure function ranges over a plurality, it is only ever defined as a cardinality function.

I change this state of affairs by treating M not as an ambient element of the grammar but as a parameter. Now, this set is like possible worlds, contexts, assignment functions, times, locations, etc. in being an element against which a logical form is evaluated. While a subtle change, this crucially grants us the ability to shift M to different sets (or different orderings of the same set) with intensional operators. Especially exciting for our purposes is that these intensional operators can be introduced in narrow syntax (compare belief predicates, for example (Von Fintel and Heim, 2011)). For M, its operator is M-SHIFT.

I define M-SHIFT as in (91), taking a predicate of individuals and returning another where the set of evaluation is M' (92) rather than the "matrix" set M. As a result, measure functions in the scope of M-SHIFT will be defined with the Bale & Barner algorithm relative to M' instead of M. While M and M' are identical in terms of their members, the ordering relation has been reversed so that cardinality is now maximal with respect to other forms of measure. Per the algorithm, variable measure functions will be non-cardinal even if they are ranging over a plurality.

(91) $[[\text{M-SHIFT}]] = \lambda P_{\langle e,t\rangle} \lambda x_e.[[P(x)]]^{M'}$

(92) $M' = \{\mu_{vol}, \ldots < \mu_{card}\}$

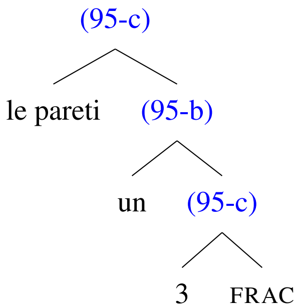
[21] See also Roberts (2025).

Let us now walk through the semantic derivation step by step. I assume fractions are functions which take two numbers and returning a two-place predicate which is true if x is part of y and the degree of x is the appropriate fraction of the degree of y. By default, the measure functions are defined per M, thus making any μ which takes a plurality a cardinality function.

(93) $[[\textsc{frac}]]^M = \lambda n_n \lambda m_n \lambda y_e \lambda x_e . x \sqsubset y \wedge \mu(x) = \mu(y) \times \frac{m}{n}$

The less-marked count reading proceeds as in (94)-(95), presumably with an additional step of universal quantification or application of a choice function to allow the structure to combine with predicates. I assume that the preposition/genitive case is semantically vacuous. Additionally, I treat indefinite articles in fractions (*un*, *a*, etc.) as the numeral one, which is synchronically plausible given the strong diachronic tendency for the numeral to be recruited for indefiniteness (Givón, 1981).

(94)



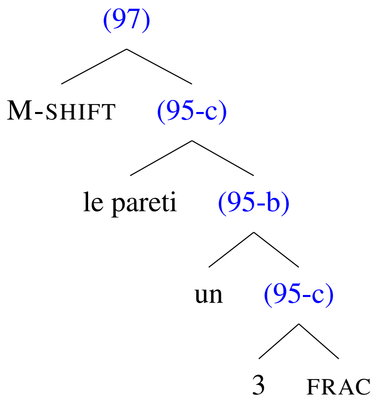
(95) a. $[[\text{terzo}]]^M = \lambda m \lambda y \lambda x . x \sqsubset y \wedge \mu(x) = \mu(y) \times \frac{m}{3}$
b. $[[\text{un terzo}]]^M = \lambda y \lambda x . x \sqsubset y \wedge \mu(x) = \mu(y) \times \frac{1}{3}$
c. $[[\text{un terzo delle pareti}]]^M = \lambda x . x \sqsubset \mathtt{max}(\lambda y . *wall(y)) \wedge |x| = |\mathtt{max}(\lambda y . *wall(y))| \times \frac{1}{3}$

The fact that there is no transitivity suspension may invoke some anxiety at this point. In the system I present, partitives can denote the subatomic parts of a plurality in addition to the plurality's members. Additionally, rational numbers like 2.5/*two and a half* are known to independently appear in natural language as nominal modifiers. Given this fact, it is fair to wonder whether cardinality functions could also return rational numbers. If we are living in the world of transitivity suspension, this is not a problem—all partitives (including FPs) which take a plural argument can only denote integrated wholes and their sums. With this gone, a cardinality function which can return non-integers would yield count FPs which can denote 3.33 apples out of 10. As as I am aware, this kind of reading is reserved only for measure FPs (not count). At face value, these facts would suggest that the analysis overgenerates for count FPs.

Closer examination would suggest this concern is unfounded, however. Natural language instances of rational numbers like 2.5 have been analyzed not as simplex numerals like *two* or *three*, but complex objects which are the sum of a cardinality operation which only returns integers and a partiality function which measures the degree of a part relative to a whole (Haida and Trinh, 2021). This dovetails neatly with the finding that languages assemble complex numerals by syntactic operations (Ionin and Matushansky, 2006); that decimal numbers appear to be composed of simplex pieces lends them to a similar analysis. Given these facts, it is reasonable to conclude that cardinality functions cannot return non-integer cardinalities on their own. Ergo, the denotation of a plural FP evaluated relative to M will exclude subatomic parts, a plurality plus a subatomic part, etc.

To get the measure reading, M-SHIFT applies to this structure and changes M to M' (96). Now, even a plurality can be measured along a non-cardinal dimension.

(96)



(97) $[\![\text{M-SHIFT un terzo delle pareti}]\!]^{M'} = \lambda x.x \sqsubset \texttt{max}(\lambda y.*wall(y)) \wedge \mu(x) = \frac{\mu(\texttt{max}(\lambda y.*wall(y)))}{3}$

There remains a loose end. In §1, I show that fraction agreement in German and Russian can have a measure or count interpretation; this distinguishes these languages from Italian, French, and Greek, which only have a measure interpretation if the fraction is targeted for agreement. If we believe that fraction agreement is the result of the M-SHIFT operator, the proposal so far undergenerates for these languages. M' as presently defined reverses the hierarchy of measure functions in M, making counting a plurality virtually impossible. This is of course a desired effect for languages like Italian, but something more needs to be said to capture German and Russian.

To resolve this issue, I posit another point of cross-linguistic variation outside of just the presence or absence of an intensional operator for measurement. I further entertain the possibility that the product of M-SHIFT's application (M') can also vary from language to language. For German, I assume that the operator shifts the set of evaluation to one where there is no ordering relation at all (98). Now, no measure function is minimal with respect to any other measure function. Consequently, all forms of measurement are equally valid as far as Bale & Barner's algorithm is concerned. This makes defining a variable measure function a purely contextually-driven process in this case, yielding both count and measure readings accessible with the application of M-SHIFT in German.

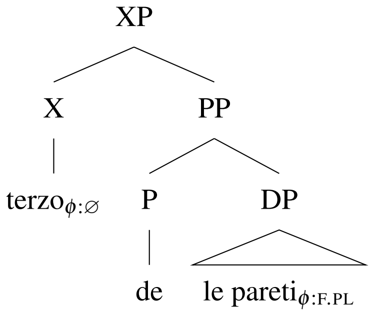
(98) $M'_{\text{German}} = \{\mu_{vol}, \mu_{card}, \ldots\}$

Russian is also a language where count and measure interpretations are possible with fraction agreement, so it is tempting to say the same thing for this language. However, recall that in §7 we saw that only the measure interpretation emerged in the presence of *čast'*. If this item is the overt realization of M-SHIFT, it is difficult to reconcile to strictness of interpretation with FP-*čast'* with looseness of interpretation for bare FPs with fraction agreement. Given this, I assume that Russian M' is like Italian (i.e., there is an ordering relation between members of the set) and M-SHIFT precludes a count reading in Russian. The possibility of a count reading with fraction agreement, I say, is contingent on other factors. This is elaborated upon in the following subsection, which details the syntactic portion of the broader analysis.

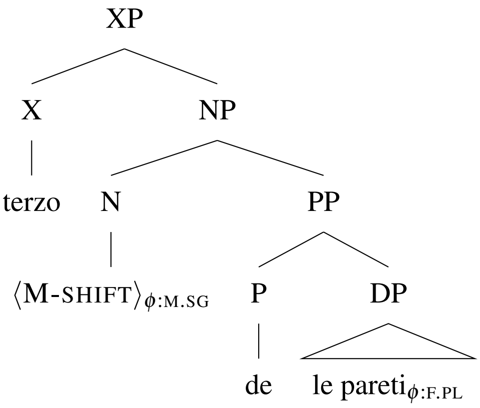
8.2 In the Syntax

An intuitive first attempt at an anlysis for the syntax of M-SHIFT may be to say that the operator itself is a kind of noun. This would allow us to assume a more-standard syntax for fraction partitives where the fraction c-commands the complement. We could assume that the structure for count FPs is that in (99), where the fraction itself is featureless. Without features on the fraction, a probe will continue to look down into the FP structure, eventually finding and copying the features on the complement. This derives complement agreement while maintaining a familiar structure for partitives (see also Klockmann 2018 on pseudopartitives). The structure for measure FPs could be like that in (100), where M-SHIFT intervenes between the fraction and the complement. Assuming that this operator comes with its own features, a probe will prefer to agree with it over the complement by locality principles. Treating M-SHIFT as a nominal with a corresponding label would also give nominal modifiers a place to attach to the structure, deriving the facts with relative clauses, adjectives, and determiners that are observed in §4.

(99) COUNT



(100) MEASURE



This kind of analysis is appealing for a number of reasons, but it leaves several questions open. To start with, an intensional operator must take scope over the variables it is shifting; if the fraction is introducing the variable measure functions, it stands to reason that M-SHIFT should outscope it at LF. The syntax in (100) derives the linear order observed when M-SHIFT is overt, but without further modification there will be a mismatch in syntactic and logical form. Additionally, supposing that fractions are featureless is at odds with the morphological facts I list in §3. As I repeat here, the denominators of many of these languages have suffixes which bear remarkable resemblance to nominal number and gender suffixes. If the locus of the features is actually M-SHIFT, a possible explanation is that the denominator is actually participating in a kind of concord with the operator. However, this offers no suitable explanation for the presence of these suffixes with count FPs. On the contrary, the structure in (99) without M-SHIFT would predict that they should not appear at all.

Finally, we see in §3 that the word for *half* French, Italian, and German is exceptionally feminine. FPs headed by *moitié* in French, for example, participate in the same agreement optionality that we see for other fractions. As a result, feminine singular agreement is possible with these FPs while masculine singular agreement (the kind seen on FPs headed by other fractions) is impossible. If M-SHIFT is the genuine target of agreement for measure FPs, then we might expect (101-c) to be acceptable. The fact that it is not invites scrutiny to the entire approach. A possible out is that French, Italian, and German has two M-SHIFT operators—one for general fractions (masculine) and one specifically for *half* (feminine)—but this is undoubtedly stipulative.

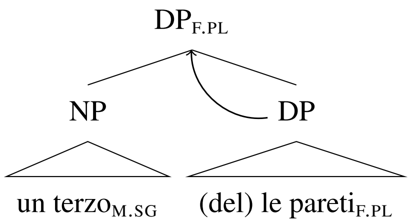
(101) 'Half the cake is moist' [French]

a.	la	moitié	des	gâteaux	est	moelleuse
	DET.F.SG	half.F.SG	of.DET.M.PL	cake.M.PL	COP.3SG	moist.F.SG
b.	la	moitié	des	gâteaux	sont	moelleux
	DET.F.SG	half.F.SG	of.DET.M.PL	cake.M.PL	COP.3PL	moist.M
c.	*la	moitié	des	gâteaux	est	moelleux
	DET.F.SG	half.F.SG	of.DET.M.PL	cake.M.PL	COP.3SG	moist.M

In lieu of the approach outlined above, I pursue an analysis where the agreement seen for measure FPs is the result of copying features from the fraction. The spirit of this story is that fractions always bear ϕ-features, but the structure of count FPs is such that the fraction is inaccessible to most outside operations. M-SHIFT changes this state of affairs by "saturating" the fraction, making it possible for Agree to target it.

Let us start with an examination of the structure for count FPs, represented in (102). Crucially, this structure is such that the fraction does not embed its complement; the fraction is merged with the complement at the DP-level, but it is the complement which projects its label (a move inspired by Selkirk 1977; Rothstein 2009). This is dependent on the assumption that genitive case/preposition seen in fraction partitives is not projecting any structure. It could be explained away perhaps as a case-marker (Chomsky, 1970) or a late-merged element (Chomsky, 1981). By virtue of projecting its label, the features of the complement percolate to the top of the structure. Since conversely the fraction does not project, the complement's features are the highest in the structure. Thus, the fraction is predicted to be the target of Agree.

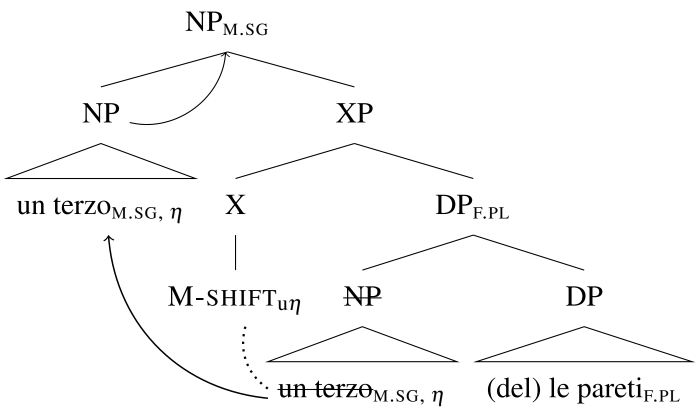
(102)



In this configuration, I will treat fractions as NPs (or some other relatively-small projection of the nominal spine). This assumption would block the presence of determiners and demonstratives in the count FP structure, a desired result given the findings in §4.[22] Treating *un*, *a*, etc. as numerals or some other projection rather than definite articles (as I do in §8.1) is once again helpful since treating them as the realization of D would block their occurrence at the left edge of count FPs. If we further assume that adjectives are hosted by one or more distinct functional projections in the nominal spine (Cinque, 2010), we also derive the impossibility of left-edge adjectives with complement agreement. There is insufficient real-estate for these items given the fraction's small size.

To derive the measure reading, M-SHIFT takes the structure in (102). In addition to it's update to the denotation, M-SHIFT comes with an unchecked feature that it needs to check—let us call this feature η. I further stipulate that fractions come with η and can thus satisfy M-SHIFT.[23] The probe on M-SHIFT searches through its c-command domain, finding the NP which bears its desired feature. After copying the features of this phrase, it will re-merge it at the top of the structure. However, this is not just any kind of movement; upon merging with the XP, the fraction will project its label (and by extension, its features) to become the top of the structure (Bhatt, 2002). The result of these syntactic acrobatics is that the fraction and its features are now more local to agreeing heads, deriving fraction agreement.

(103)



It stands to reason that the probe on M-SHIFT also copies retains a copy of the fractions ϕ-features after it has moved. In languages where M-SHIFT is overt, these features can be spelled-out—this is what appears to happen with Russian *čast'* FPs. In effect, fractions are not participating in concord with the operator; rather, the operator is participating in concord with them. The *-ja* suffix we see on the denominator in such FPs (glossed as ADJ) could perhaps be creatively reanalyzed as the product of goal-flagging in a system like Deal (2015).

The syntactic reorganization spurred by M-SHIFT makes a number of positive predictions for the observa-

[22]However, we see in (101) that some fractions in some languages permit complement agreement even with a left-edge definite determiner. I tentatively suggest that some count FPs could keep have a syntactic object as large as a DP in their left branch.

[23]This has the added benefit of restricting M-SHIFT's domain of application to only FPs. Without this or something similar, the operator could in theory apply to any predicate of individuals.

tions in §4. Firstly, now that the fraction NP is at the top of the structure, it can be taken by additional functional projects (AdjP, DP, etc.) which can take the items the count structure could not (adjectives, determiners, etc.). This happily explains the co-occurrence of these items with mandatory fraction agreement. Secondly, the evacuation of the fraction to a higher projection opens up movement possibilities of the internal DP. Assuming that movement targets the maximal projection of a given head, displacement of the complement as shown for German and Greek should be impossible with the count FP structure because the fraction is included in the maximal projection of D. Trying to move that DP should drag the fraction along with the complemment. With the fraction moved, the complement DP can now move independently of the fraction. This accounts for the impossibility of PP-displacement with complement agreement.

Also worth mentioning is that this story generates some of the agreement facts with singular complements seen in §3. There is nothing which would block M-SHIFT from applying to a singular-containing FP, and the same movement processes would apply. The main difference is that the operator's semantic contribution is vacuous if the complement is singular because (with the exception of German) M-SHIFT makes counting impossible but singularities generally cannot be counted to begin with. Regardless, this does not block the application of M-SHIFT, yielding agreement with the fraction in a singular FP.[24]

Let us conclude with with addressing the issue left open at the end of §8.1. As I say there, agreement with the fraction of Russian bare FPs yields both measure and count readings, whereas with *čast'* only the measure reading is available. This is hard to make sense of if *čast'* and M-SHIFT are the same underlying item. I would like to suggest that the count reading originates from the same syntactic structure in all cases, but the target of agreement in this structure is genuinely optional. This is motivated by an independent fact of Russian that DPs with numeral modifiers can trigger either plural or neuter singular (default number/gender) on the verb.

(104) Na kurs zapizal-{is'/os'} dvadcat' dva studenta
on course registered-{PL/3SG.N} twenty two student.PL
'Twenty-two students registered for the class' (Adapted from Ivlieva and Podobryaev 2025)

This alternation has been written about extensively and naturally has characteristics which are too in-depth to cover here (see Franks 1994; Pereltsvaig and Franks 2004; Pesetsky 1982; Ivlieva and Podobryaev 2025 and works cited therein). However, as a stop-gap suggestion to the problem the Russian data poses, we may wonder whether the structure for count FPs is amenable to the same rules which derive the agreement optionality in (104). Perhaps, given some structural closeness, Russian probes can genuinely pick two bundles features from the structure. This of course warrants an explanation of the apparent lack of flexibility in agreement we would get from the representation of the measure FP, though. If the structural closeness of two feature bundles is the cause of the optional agreement, perhaps M-SHIFT creates a sufficiently-wide structural rift to mandate agreement with the fraction's features. The efficacy of such an approach must wait for future work.

9 Conclusion

In short, this work provides an account for an agreement-interpretation alternation identified in several Indo-European languages and attempts to explain parameters of cross-linguistic variation wherever possible. I begin with the observation that fraction partitives in Italian, Greek, French, Russian, and German have two readings which I call measure and count. A novel contribution to the description of these items, it is resported that these readings can be disambiguated with certain agreement configurations; agreeing with the features of the FP's complement mandates a count reading, while copying the features of the fraction permits at least a measure reading. Agreement has a number of other syntactic correlates, like the ability for an FP to take nominal modifiers, PP-extrapose, etc. The facts in part lend themselves to established analyses such as semantic agreement, semantic type ambiguity, or the application of a semantic type-shifter, but I present a series of arguments for each showing that they do not provide a comprehensive account.

Ultimately, I propose that measure and count FPs can be resolved entirely to the presence or absence of an item I dub M-SHIFT. I take the count structure to be the unmarked case given the measure reading does not exist in some languages (American English). Syntactically, M-SHIFT takes the count FP and manipulates the

[24] However, this overgenerates for Italian (complement agreement is possible if the complement is singular) and undergenerates for English (singular-containing FPs in English can have fraction or complement agreement). Something more has to be said about both cases, but it is beyond the scope of the present study.

structure to make the features of the fraction accessible to agreement. Semantically, M-SHIFT is an intensional operator which changes a parameter by which variable measure functions are defined.

This analysis has a number of advantages over its competitors in handling the reported facts. It can generate both readings regardless of syntactic position and regardless of the grammatical number of the fraction. It takes into account the differences in syntactic behavior depending on what agreement is licensed. It also maintains a simple ontology for pluralities and permits a measure reading whose denotation can be a plurality (not just a mass). All of this is done with, in my view, relatively minimal additional assumptions to established perspectives on the syntax-semantics interface.

This is at the expense of one very drastic change, though, and that is how measure functions are defined. As stated previously, defining a function for μ is taken to be a process determined by context and the ontological status of μ's argument (i.e., whether it is singular or plural). This work is a radical deviation to these assumptions in (i) treating the set of possible measure functions as a parameter which can be shifted, and (ii) introducing intensional operators in narrow syntax to shift them. As far as I know, there are no prior suggestions of this type in the literature. Skepticism is a warranted reaction to this analytical move; while it outperforms alternatives in some ways, it is fair to wonder if this is the best answer given its degree of specialization and, conversely, its lack of generalizability. For this analysis to hold water, it seems necessary to identify other phenomena which (i) mandate other forms of measurement over counting, and (ii) are necessarily products of elements introduced in narrow syntax (as opposed to, e.g., the context). Whether or not these items exist is a question for future work.

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